WEEK 10

# Data Science III

Does course context transfer? Comparing general and context-aware risk models

Bera Ergin · s29315 · PJAIT · Türksat internship

# Why this experiment?

*In real life, risk may depend on the course, the teacher, even the teacher's approach in a given year. A student who is safe in one course with certain grades may be at risk in another course with a hard final.*

**If so, a general model that looks only at the student will be wrong, and a context-aware model should do better. This week we measured that.**

## What can OULAD test?

**Course** — Yes

Module identity and past runs of the course

**Term** — Yes

Presentation: a rough proxy for "that year's conditions"

**Teacher** — No

Not in the dataset

*The teacher dimension could not be tested; it moves to future work.*

# Experiment design: three arms, two extra checks

| Set | What it knows | Question it asks |
|---|---|---|
| `GENERAL` | The student's assessment and activity trace | Baseline |
| `GENERAL_MODULE` | + course identity | Is knowing the course enough? |
| `HISTORY` | + past runs of the course (assessment difficulty, risk rate) | Does knowing the course's past help? |
| `HISTORY_SLOT *` | HISTORY without the past risk rate | Is the problem a single column? |
| `GENERAL_REL *` | Position within the class (rank percentiles) | Does context transfer as within-class rank? |

Tuning: train on 2013J → select on 2014B (log loss) · Single evaluation: 2014J · 1,000 stratified bootstrap samples · Two model types: logistic regression, gradient boosting (hgb)

** Added after the first test results were seen (exploratory). The test set was therefore frozen and the validation period reported separately.*

# First result: course history did not improve prediction

| Set | hgb ROC AUC [95%] | logreg | Brier (hgb) |
|---|---|---|---|
| GENERAL | 0.792 [0.783–0.801] | 0.786 | 0.177 |
| GENERAL_MODULE | 0.799 [0.790–0.808] | 0.805 | 0.175 |
| HISTORY | 0.783 [0.773–0.792] | 0.773 | 0.190 |
| HISTORY_SLOT | 0.783 [0.774–0.793] | 0.775 | 0.190 |
| GENERAL_REL | 0.804 [0.795–0.813] | 0.792 | 0.169 |

**HISTORY − GENERAL**

hgb

**−0.010 [−0.012; −0.007]**

logreg

**−0.013 [−0.015; −0.012]**

Paired bootstrap intervals exclude 0: the gap is not noise. Brier also gets worse.

# Bug or finding? We questioned the code first

## Assessment matching

Assessments come in the same order across runs, only shifted by a few days.

**Code is correct**

## Past risk rate

Only 8 distinct values in train: the model learns this column from 8 presentations, not 13,639 rows.

**Weak signal**

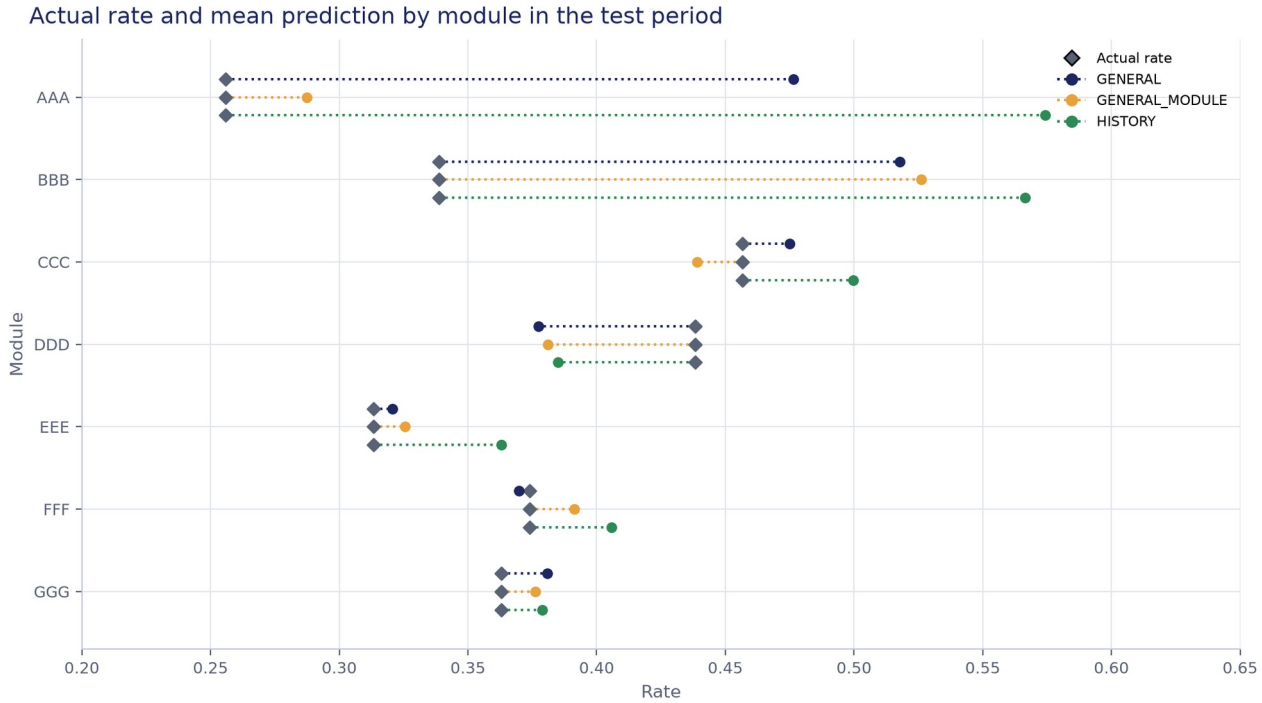
## Drift over time

Risk is 42.6% in train, 38.3% in test. In BBB the past rate is 0.42; the actual test rate is 0.339.

**The past stays high**

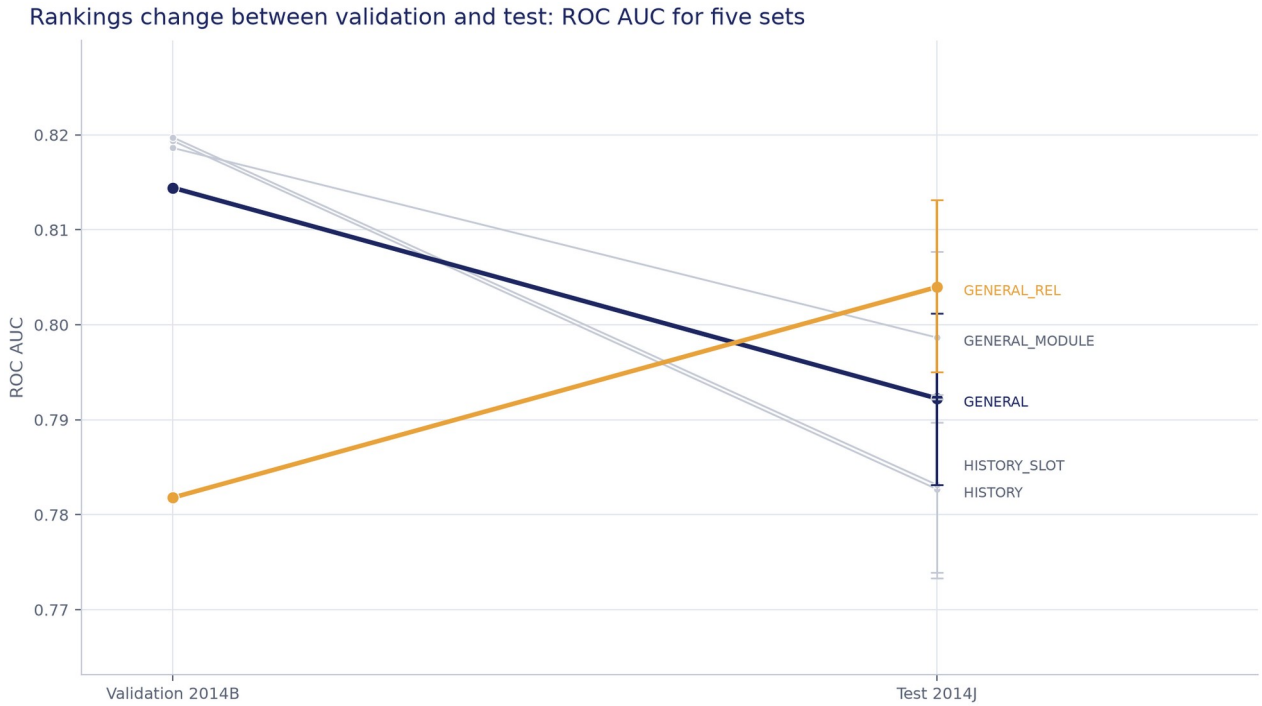
*Conclusion: no code bug, the finding is real. But the explanation was still incomplete; the next steps tested it.*

# Context exists in the base rate, but it drifts between terms




- In AAA the actual rate is 0.256: GENERAL predicts 0.476, HISTORY 0.574; course identity (0.287) corrects it.
- In BBB all three sets predict between 0.52 and 0.57; actual is 0.339.
- The past risk rate pushes predictions up in the test period.
- Decision: the system does not show absolute probabilities.

# The set leading on test comes last on validation




- GENERAL_REL is highest on test (0.804) and lowest on validation (0.782).
- This set was designed after looking at test results; its test gain is exploratory.
- Decision: GENERAL_REL was not adopted; the test set was frozen.
- Lesson: designing against the test set can pick a lucky winner.

# Within a class, the sets are almost identical

| Set | Test 2014J | Validation 2014B |
|---|---|---|
| GENERAL | 0.796 | 0.810 |
| GENERAL_MODULE | 0.797 | 0.811 |
| HISTORY | 0.793 | 0.811 |
| HISTORY_SLOT | 0.794 | 0.811 |
| GENERAL_REL | 0.800 | 0.795 |

*Within-module ROC AUC, unweighted mean across modules (hgb).*

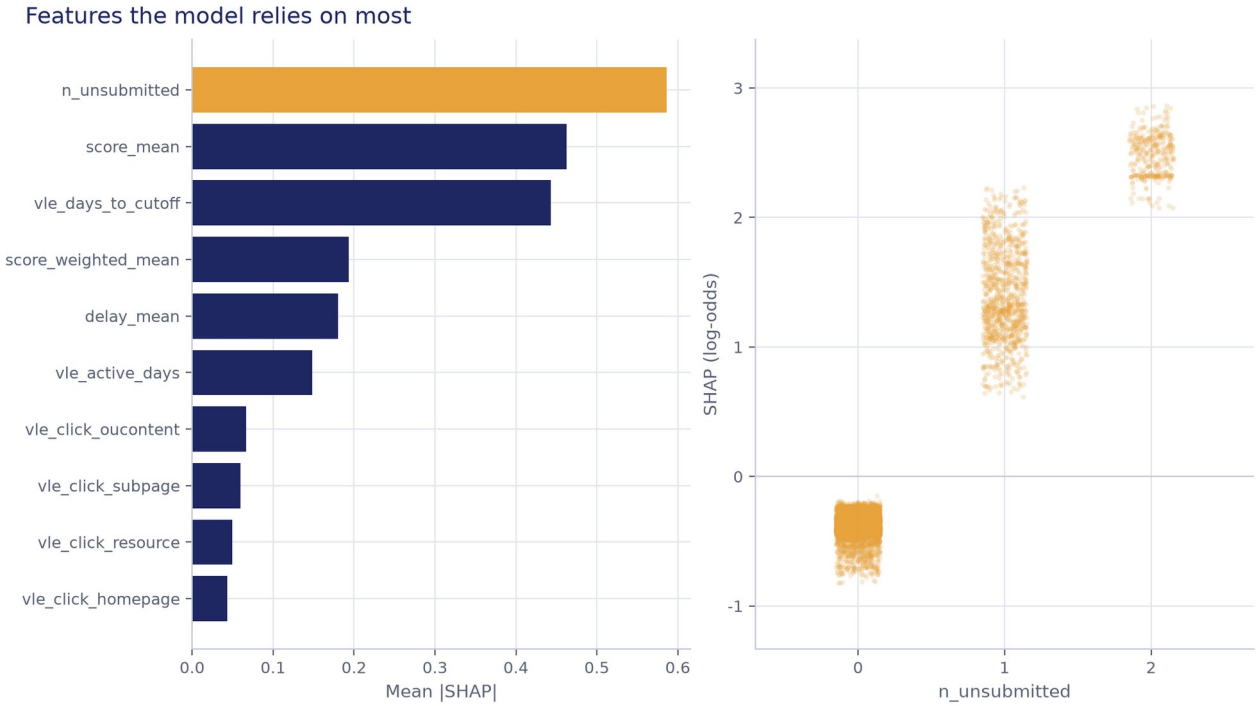
- Differences are a few thousandths; the ordering changes between terms.
- No context recipe consistently helped identify who is at risk in both terms.
- Context matters at the level between courses (base rate), not in within-class ranking.

# Unseen course: identity does not transfer, a course without assessments breaks



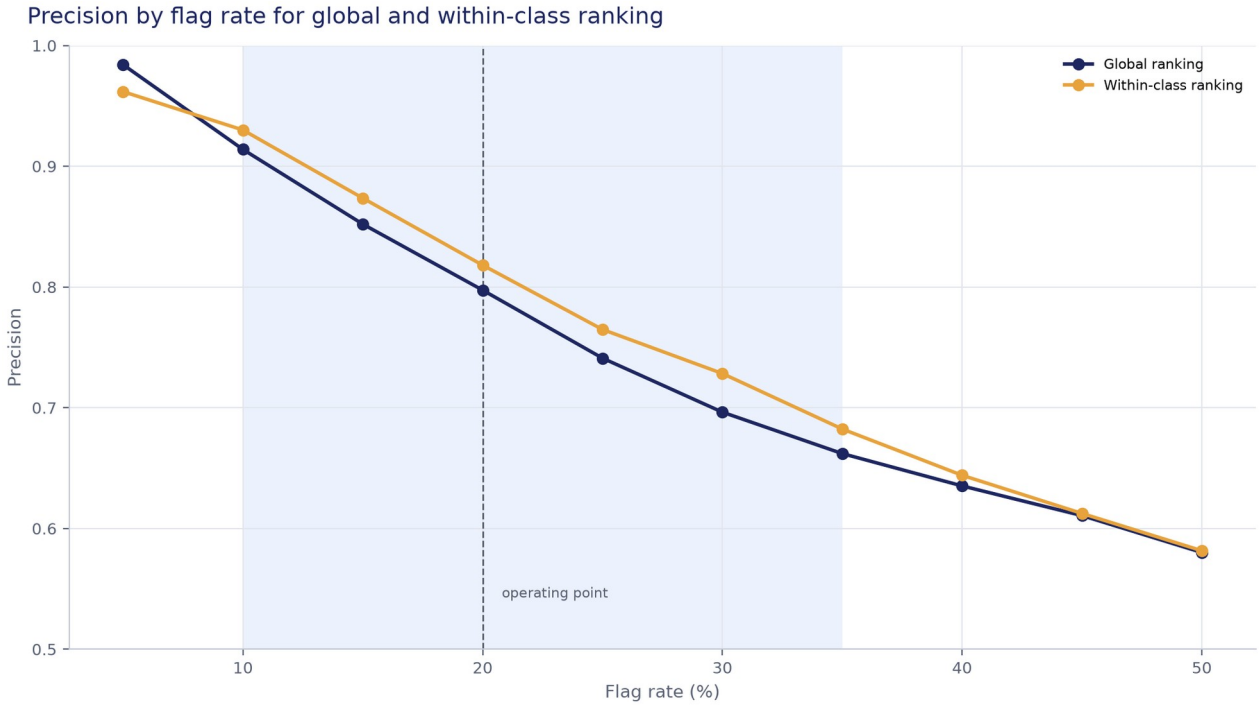

- Each course was left out of training in turn (LOMO). Mean ROC AUC: GENERAL 0.784, MODULE 0.785, REL 0.787, HISTORY 0.782.
- Differences are small and no intervals were computed.
- GGG: prediction ~0.93, actual 0.36. The model reads "no assessment yet" as "did not submit".
- Rule: no flag in a course with no exam yet.

# The model relies most on submissions, grades and recent activity




- Mean |SHAP| (log-odds): n_unsubmitted 0.586 · score_mean 0.462 · vle_days_to_cutoff 0.443.
- School counterparts: missed exam, grade average, attendance.
- SHAP describes model behaviour. It does not mean "not submitting causes risk".

# Decision rule: top 20% in each class




- Precision at 20%: global 0.797, within-class 0.818.
- Within-class ranking leads between 10% and 35%; global leads at 5% (0.984 / 0.962); equal from 45%.
- Within-class rank is not affected by base-rate drift and gives each teacher equal follow-up capacity.
- Cost: lower precision in low-risk classes (AAA 0.62).

# Bridge to the system: transparent rule vs model

### Rule (fixed before seeing results)

- Rank percentiles of 3 signals per class: unsubmitted assessments, grade average (reversed), active days (reversed)
- Mean → top 20% within class
- No flag in a course without a graded assessment
- No training; at_risk not used

| Term | Rule | Model (hgb) | Hits rule / model |
|---|---|---|---|
| 2014B | **0.900** | 0.934 | 975 / 1,012 |
| 2014J | **0.787** | 0.829 | 1,283 / 1,352 |

*Precision of the within-class 20% flag; GGG excluded in both terms.*

- The rule captures 95–96% of the model's hits.
- The model leads in 11 of 11 module-terms (ROC AUC gap 0.001–0.029).
- The school system has no outcome labels to train on: a model was never an option. We measured the cost of transparency.

# Decision mechanism: each decision rests on evidence

| Decision | Evidence |
|---|---|
| **HISTORY rejected** | Paired difference negative for both model types; interval excludes 0 |
| **GENERAL_REL not adopted** | Lowest on validation; test gain is exploratory |
| **Reference model: GENERAL hgb** | Simple, stable across both terms, transferable to any course |
| **No probabilities shown** | The base rate drifts between terms |
| **Top 20% within class** | Capacity curve: within-class leads between 10% and 35% |
| **Transparent 3-signal rule in the system** | No labels at school; cost ~4 points of precision |
| **No flag in a course without exams** | GGG: ~0.93 predicted in an unseen course |

# Answer to the hypothesis

## Supported

- Risk differs a lot between courses, and the difference is real.
- Course identity improves calibration and overall ranking on test.

## Not supported

- Knowing the course's past does not rank students more accurately within their class.
- The past risk rate drifts over time and is learned from few terms.

## Not testable

- The teacher and the teacher's approach that year (not in OULAD).
- Testable on school data as outcome labels accumulate.

**In short: risk does depend on the course, but this information does not transfer from the course's past to the student. What works is judging a student within their own class.**

# Explanation log: two wrong guesses, two corrections

| Guess | Check | Result |
|---|---|---|
| The drop comes only from the past risk rate | That column was removed (HISTORY_SLOT): result unchanged | **Wrong: past assessment features do not generalise either** |
| GENERAL_REL only misses the level; its ranking is sound | Ranking (ROC AUC) measured on validation: lowest | **Wrong: the set was not adopted** |

> *No unverified explanation was reported as a result; each guess was tested by a check.*

# Bug record 1: figure alternative text

| Field | Description |
|---|---|
| **Symptom** | The alt text of the unseen-course figure described the highlighted row as "the course left out of training". |
| **Root cause** | In this analysis every row is a course unseen in training. GGG is highlighted for another reason: it has no graded assessment by day 60. |
| **Fix** | Title, alt text (TR/EN) and an explanatory note on the GGG row were updated. |
| **Status** | Verified: figures.json and the regenerated figure were checked. |

# Notebook and the single-source principle

| | |
|---|---|
| `03_oulad_model.ipynb` | Turkish |
| `03_oulad_model_eng.ipynb` | English |
| `notebooks/html/` | HTML exports |

No model, metric or plotting code is written in the notebook; everything is called from the data_science/oulad package. The same PNGs are used in the notebook, this deck and the weekly report.

a. Question
b. Design
c. Test results
**d. Two terms**
e. Base-rate drift
f. Unseen course
g. SHAP
h. Decision rule
**i. Bridge to the system**
j. Limitations
k. Summary

# Verification

**266 passed**

1 warning (StarletteDeprecationWarning)

```
python -m data_science.oulad.model --lomo
python -m data_science.oulad.explain
python -m data_science.oulad.figures
python -m data_science.oulad.rule
```

- Notebook 03 (TR/EN) executed from scratch in the user's terminal: 3 min 42 s, 0 error cells.
- HTML exported after the notebooks; no alt text warning.
- The OZET cell matches the raw outputs exactly.
- Tests: no leakage in sets, tuning never sees test, paired difference and bootstrap deterministic, rule never uses at_risk.

# Open items and Week 11

## Open items

- n_unsubmitted ↔️ "missed exam": to verify in the school schema (the rule's first signal).
- pytest warning (Starlette/httpx) not resolved.
- Alternative text: solved in 03; open in 01 and 02.
- Week 6 bug 4, Week 7 m5 and c4 open.

## Week 11 — Risk integration

- Risk tab = transparent rule: top 20% within class, with reasons
- No flag in a course without exams; no probabilities
- get_risk tool for the assistant (respects RLS, no invented causes)
- Data Science tab, tests, Docker, security